SAMAR FATHALAH

Machine Learning & Computer Vision | Software Engineering

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Summary

Computer engineering graduate with an ***M.Sc. in Information Technology***, specializing in ***machine learning and generative AI***. Hands-on experience applying data augmentation and semantic segmentation to enhance the robustness of deep learning models for computer vision. Backed by ***2+ years*** of production **software engineering** in Python, APIs, and AWS.

Experience

Research Assistant — Master's Thesis | University of Stuttgart, Germany | 09/2025 - 03/2026 | Slides

Thesis on improving the safety of ***end-to-end autonomous driving models*** in rare, safety-critical scenarios.

- Developed a ***diffusion-based data augmentation pipeline*** in **PyTorch** to synthesize collision and near-collision scenarios from the ***nuScenes*** dataset by injecting adversarial vehicles, generating safety-critical cases underrepresented in real-world driving data.
- Fine-tuned planning-oriented end-to-end autonomous driving models on the augmented dataset, targeting improved decision-making in safety-critical scenarios.
- Evaluated performance through ***open-loop and closed-loop*** frameworks, reducing the collision rate by ***39%.***

Working Student | Valeo Schalter und Sensoren - Bietigheim-Bissingen, Germany | 01/2025 - 06/2026

- Built and containerized ***Flask and FastAPI REST*** services with ***Docker*** to access and deliver autonomous-driving test data from MySQL databases, including sensor recordings, calibration data, and vehicle metadata.
- Developed user/client dashboards using ***Grafana*** to ***visualize and analyze*** sensor data, e.g., RADAR, LiDAR, and vehicle information from autonomous-driving test environments.
- Automated ***sensor-system integration*** through driver installation, system configuration, and ***Python/C-based*** tooling.

Full Stack Developer | Khazenly - Cairo, Egypt | 01/2023 - 09/2024

- Built merchant-facing features for a ***self-service fulfillment platform*** serving ***250+ merchants***, enabling automated order fulfillment and real-time order tracking without operations-team involvement.
- Scaled automated order fulfillment flows between merchants, customers, and logistics partners using ***AWS (Lambda, S3, SQS)***, improving delivery reliability and handling more than ***700 orders/day.***

Teaching Assistant | The German University in Cairo - Cairo, Egypt | 08/2022 - 01/2023

- Designed course content, technical projects, and assessment materials for the "***Introduction to Media Engineering"*** course for third-year students in the ***Media Engineering and Technology*** major.

Skills & Languages

- **Languages:** English (C2), German (A2, actively improving).
- **Programming Languages:** Python, Java, JavaScript, C/C++, C#, Golang.
- **Machine Learning Techniques:** Convolutional Neural Networks (CNNs), Transformers, Generative Adversarial Networks (GANs), Diffusion Models, Long Short-Term Memory networks (LSTM), Deep Neural Networks (DNNs), Semantic Segmentation, Fine-tuning, Variational Autoencoders (VAE), Attention Mechanisms, Self-supervised learning.
- **Data Engineering:** Data preprocessing, Augmentation, Feature extraction, Model evaluation, and Validation.
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Hugging Face, scikit-learn, OpenCV, NumPy, Pandas.
- **MLOps & Infrastructure:** Docker, AWS (Lambda, S3, EC2), Git, GitHub, CI/CD.

Education

M.Sc. Information Technology (Software and Hardware Engineering) | University of Stuttgart (2024 - 2026) | GPA (2.00)

B.Sc. Media Engineering & Technology (Computer Science) | The German University in Cairo (2017- 2022) | GPA (2.2)

Projects

Master's Research Project | ***Semantic Image Synthesis under Adverse Weather Conditions*** | ***Slides***

- Built an extended ***DP-GAN framework*** for generating ***adverse-weather images***, guided by semantic segmentation maps and clear-weather images.
- Proposed a ***novel fusion strategy*** integrating clear-weather features with semantic maps via a three-pyramid generator architecture using a learnable fusion mechanism.
- Validated the synthesized images with a downstream ***semantic segmentation model,*** achieving an ***FID score of 98.01***.

Bachelor Thesis | ***Automatic Music Transcription (Audio to Sheet Music)***

- Built an audio-to-sheet-music transcription system to convert musical notes into readable sheet music.
- Compared multiple ***machine learning models (DNN, CNN, LSTM)***, with the CNN achieving the best performance at ***75%*** average precision.